
Solution to Exercise Sheet 4:
Tensors in probability theory

Foundations of Neuro-Symbolic AI
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Task 1:

The probability distribution $\mathbb{P}[X_0, X_1, X_2]$ is a tensor of order 3:

$$\begin{pmatrix} 0.648 & 0.015 \\ 0.072 & 0.005 \end{pmatrix} \begin{pmatrix} 0.072 & 0.135 \\ 0.008 & 0.045 \end{pmatrix} [X_0, X_1, X_2]$$

(i) The marginal distribution of *Traffic* (X_0) and *Crash* (X_1) is the contraction

$$\mathbb{P}[X_0, X_1] = \langle \mathbb{P}[X_0, X_1, X_2] \rangle_{[X_2]} = \begin{pmatrix} 0.72 & 0.15 \\ 0.08 & 0.05 \end{pmatrix} [X_0, X_1] .$$

(ii) The distribution of *Crash* (X_1) conditioned on *Traffic* (X_0) is the normalization

$$\begin{aligned} \langle \mathbb{P}[X_0, X_1] \rangle_{[X_0]} &= \begin{pmatrix} \frac{72}{72+15} & \frac{15}{72+15} \\ \frac{8}{8+5} & \frac{5}{8+5} \end{pmatrix} [X_0, X_1] = \begin{pmatrix} \frac{24}{13} & \frac{5}{13} \\ \frac{8}{13} & \frac{5}{13} \end{pmatrix} [X_0, X_1] \\ &\approx \begin{pmatrix} 0.828 & 0.172 \\ 0.615 & 0.385 \end{pmatrix} [X_0, X_1] . \end{aligned}$$

Task 2:

The event corresponds with the propositional formula

$$f[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1] .$$

To compute the probability that this event holds we perform the contraction

$$\begin{aligned} \langle \mathbb{P}[X_0, X_1, X_2], f[X_0, X_1] \rangle_{[\emptyset]} &= \langle \mathbb{P}[X_0, X_1], f[X_0, X_1] \rangle_{[\emptyset]} \\ &= 0.72 + 0.15 + 0.05 \\ &= 0.92 . \end{aligned}$$